Human Activity Classification Based on Micro-Doppler Signatures Using a Support Vector Machine

Youngwook Kim, *Member, IEEE*, and Hao Ling, *Fellow, IEEE*

***Abstract*—The feasibility of classifying different human activities based on micro-Doppler signatures is investigated. Measured data of 12 human subjects performing seven different activities are collected using a Doppler radar. The seven activities include running, walking, walking while holding a stick, crawling, boxing while moving forward, boxing while standing in place, and sitting still. Six features are extracted from the Doppler spectrogram. A support vector machine (SVM) is then trained using the measurement features to classify the activities. A multiclass classification is implemented using a decision-tree structure. Optimal parameters for the SVM are found through a fourfold cross-validation. The resulting classification accuracy is found to be more than 90%. The potentials of classifying human activities over extended time duration, through wall, and at oblique angles with respect to the radar are also investigated and discussed.**

***Index Terms*—Human activity classification, micro-doppler, support vector machine, through-wall.**

I. Introduction

DUE TO THE increasing demand for physical security and surveillance, one of the emerging applications of radar recently is the detection and tracking of people in a highly cluttered environment [1]–[11]. For instance, it is desirable to locate humans through walls and extract information on their activities in real time by using electromagnetic-based sensors. The applications of through-wall human monitoring include disaster search-and-rescue, physical security, law enforcement, and urban military operations.

There has been a significant amount of research for developing human detection and tracking sensors. One approach is the use of wide bandwidth (BW) to achieve high-resolution imaging in the down-range dimension. Several studies have shown promising results from imaging building interiors to ultimately detecting and tracking humans in indoor environments as reported in [2]–[8]. Another choice is to use a Doppler radar [1], [9]–[22]. Doppler-based radars are excellent for detecting movement while suppressing any stationary clutters in the background. Low-cost sensors using off-the-shelf components or even single-chip modules are readily available. Finally, a very unique and interesting aspect of Doppler returns from humans is the appearance of "micro-Doppler" features [9], [12]. Micro-Dopplers are generated from the non-rigid-body motions of humans and contain valuable information related to human movements. Since it was first reported in [9], a number of works have appeared on exploiting micro-Dopplers for human classification. For example, van Dorp [13] estimated the parameters of human gait from frequency modulated continuous wave radar data. Otero [14] designed a simple classifier to recognize walking humans using a spectral analysis. In [15]–[17], micro-Doppler features of radar target returns were extracted through various time-frequency analyses. Micro-Doppler features were explored to distinguish among humans, animals, and vehicles in [18]–[22]. In this paper, we pose a different research question, namely, whether it is possible to distinguish various human activities such as walking, running, crawling, etc., based on the micro-Doppler features. The reliable recognition of simple human activities may be a first step toward addressing the more difficult yet important question of determining human intent.

Our approach is to apply a support vector machine (SVM), trained using measured micro-Doppler data, to classify activities of a human. SVM is a binary classifier that constructs a maximal-margin hyperplane to separate data between classes [23]–[25]. It has been used extensively for many diverse classification problems due to its superior performance over other classification methods such as the Fisher linear discriminator and the Bayesian decision method. In the radar community, SVM has been applied to array signal processing [26], [27] and radar target recognition [28], [29] problems. In this paper, we consider seven different human activities such as running, walking, walking while holding a stick, crawling, boxing while moving forward, boxing while standing in place, and sitting still. In order to recognize the activities, the time-varying Doppler signatures are examined using the short-time Fourier transform (STFT), and different features are extracted from the spectrogram. Based on the extracted features, an SVM is constructed to classify the seven activities. For the generation of training data, measurements on 12 human subjects are collected in the laboratory using a 2.4-GHz Doppler radar [10]. To implement a multiclass problem using the SVM, which is a binary classifier, a decision-tree-based structure is

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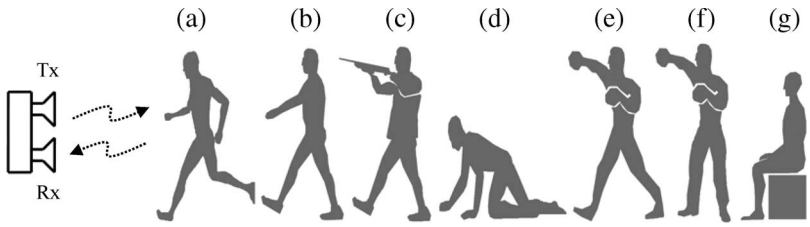


Fig. 1. Illustration of measurements of seven different human activities.

employed. The training process and the resulting classification accuracy are evaluated. The importance of each feature is also investigated. Finally, the potentials of classifying human activities over extended time duration, through wall, and at oblique angles with respect to the radar are investigated and discussed.

This paper is organized as follows. The measurement data collection of the different human activities is described in Section II. Section III discusses the feature selection and extraction process from the spectrogram. In Section IV, the concept and training results of the SVM are described. Section V addresses some additional issues of monitoring humans including through-wall. Section VI presents the conclusion.

II. Measurement of Different Human Activities

Measured data of 12 human subjects undergoing different activities are collected using a Doppler radar testbed [10]. The radar operates at 2.4 GHz. The transmitter operates in the continuous-wave mode. The receiver comprises a microstrip antenna connected to a commercial integrated receiver board. The antennas are installed to collect vertical polarization data. Since the human body is typically vertically elongated, the use of vertical polarization results in a stronger radar return. The received signals are downconverted and then digitized (at 1 kHz) for offline processing. The measurements are performed in an indoor environment under the line-of-sight condition. Only a single human subject is tested at one time, with the subject moving directly toward the radar. The range of the measurement is between 2 and 8 m.

Data for human subjects undergoing seven different activities are collected, such as follows: (a) running, (b) walking, (c) walking while holding a stick (using both hands), (d) crawling, (e) boxing while moving forward, (f) boxing while standing in place, and (g) sitting still (with slight fidgeting movements). An illustration of the performed activities is shown in Fig. 1. The descriptions of each activity are given in Table I. The data are collected over a 6-s interval. The STFT is then applied to the data to generate the corresponding spectrogram. A 0.25-s Kaiser window is used in the STFT. The resulting spectrograms of the seven activities from one particular subject are shown in Fig. 2.

As shown in Fig. 2, the spectrograms show rather interesting and distinct signatures depending on the activity. The strongest return in each spectrogram comes from that of the torso, while the periodic micro-Doppler modulations surrounding the torso return come from arm and leg movements. Among all the activities, the running motion has the highest Doppler frequency due to the torso, which can be as high as 30 Hz. It also has a wide Doppler signal spread, which can be as large as 120 Hz. The walking motion shows similar Doppler patterns with the running case, except that the maximum Doppler shift from the torso is lower (20 Hz) and the period of the micro-Doppler is longer. Walking while holding a stick has a similar spectrogram with the walking case, but it has a slightly narrower micro-Doppler frequency spread. In the crawling motion, the torso Doppler is nearly zero. In addition, the spectrogram shows that most of the micro-Dopplers are skewed toward the positive with respect to the torso Doppler. This is expected since contrary to the earlier motions, there is no back-swing in the crawling motion. In the boxing while standing in place motion in Fig. 2(f), periodic micro-Dopplers from the arms are clearly observed. In Fig. 2(e), the boxing while moving forward motion has a positive torso Doppler component in addition to the micro-Dopplers from the arms. In comparison to normal walking, it is also clear that the micro-Doppler spread is quite wide due to the rapid punching motion. The sitting still has almost a zero torso frequency and shows small sporadic micro-Dopplers due to fidgeting motions. These characteristics of Doppler signatures will be extracted in the next section.

TABLE I
Seven Human Activities Under Study

Activity	Description
(a). Running	The act of running forward quickly by moving arms and legs.
(b). Walking	The act of walking forward at a moderate speed while moving arms and legs.
(c). Walking while holding a stick	The act of walking while holding a wooden stick in a horizontal position using both hands.
(d). Crawling	The act of crawling on both hands and knees while moving forward on the ground.
(e). Boxing while moving forward	The act of throwing punches using both arms while walking forward.
(f). Boxing while standing in place	The act of throwing punches using both arms while standing still.
(g). Sitting still	The act of sitting in a chair with slight fidgeting movements such as shaking of legs, touching of hair, or crossing of arms.

III. Feature Extraction and Training Data Generation

A. Doppler Signal Detection and Feature Extraction

In order to extract distinct Doppler features from a spectrogram, it is necessary to distinguish the Doppler signal from noise. A key issue is the determination of a noise threshold. A histogram of the signal strength of the noise (i.e., received signal with no target present) is shown in Fig. 3(a). In the figure, the noise is found to have a Gaussian-like distribution. Fig. 3(b)

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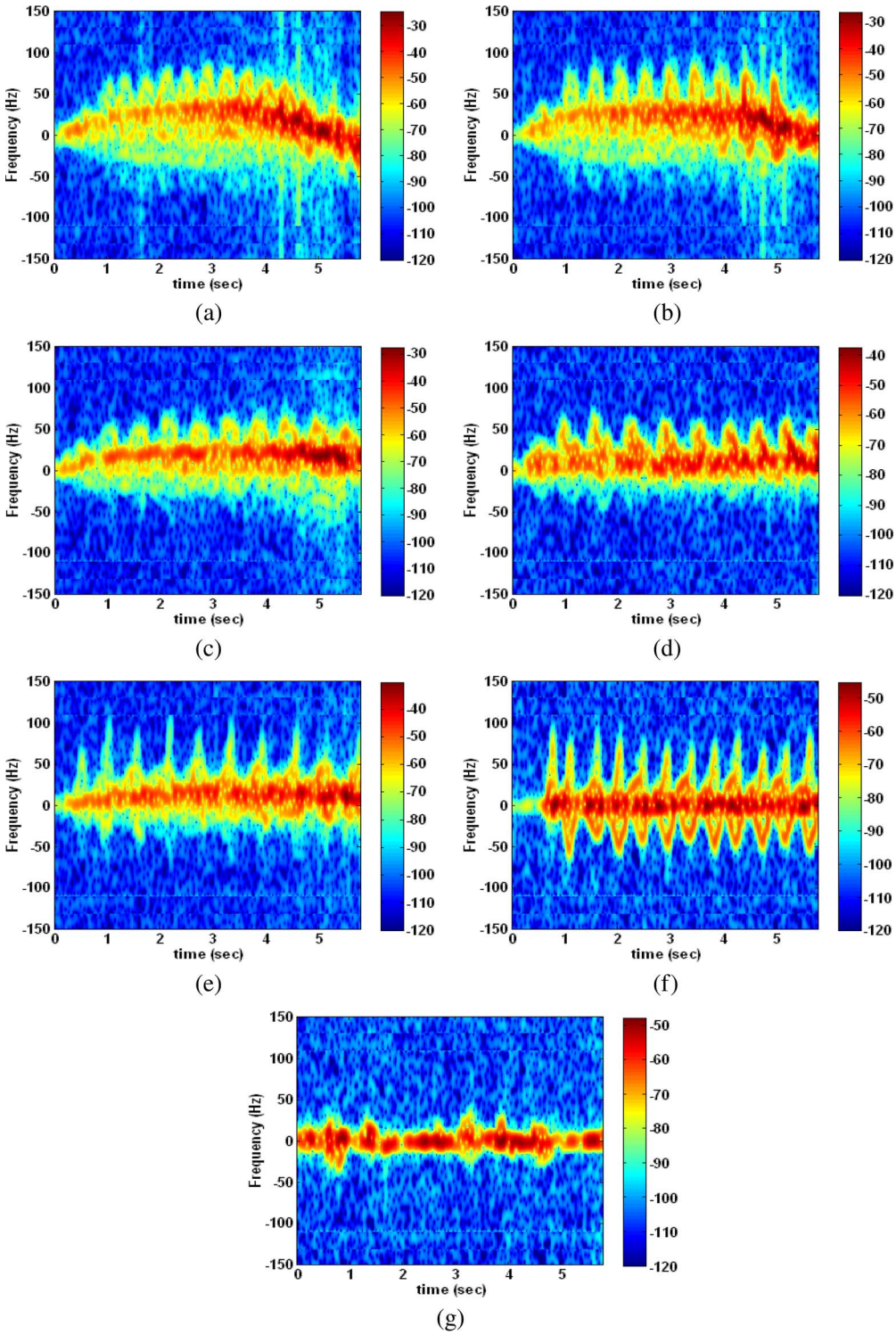


Fig. 2. Spectrograms of seven human activities. (a) Running. (b) Walking. (c) Walking while holding a stick. (d) Crawling. (e) Boxing while moving forward. (f) Boxing while standing in place. (g) Sitting with slight movements.

shows a histogram when the human Doppler signal is present. Thus, the lowest power level at which the signal histogram starts to deviate from the Gaussian-like noise distribution can be used as the noise threshold, which, reading from the figure, is −83 dBm. By using this threshold, the spectrogram of a representative activity (boxing while standing in place) is processed and shown in Fig. 4. We note here that while this approach works well in the present line-of-sight situation, it may not work in more complex environments containing interference, through-wall, or multipath effects.

Next, different Doppler features are extracted from the denoised spectrogram. The spectrogram itself can be used as input to a classifier without any processing to identify different activities. However, such high-dimensional data would result in a classifier with a very complex inner structure and, thus, an enormous training process. The proper choice of features

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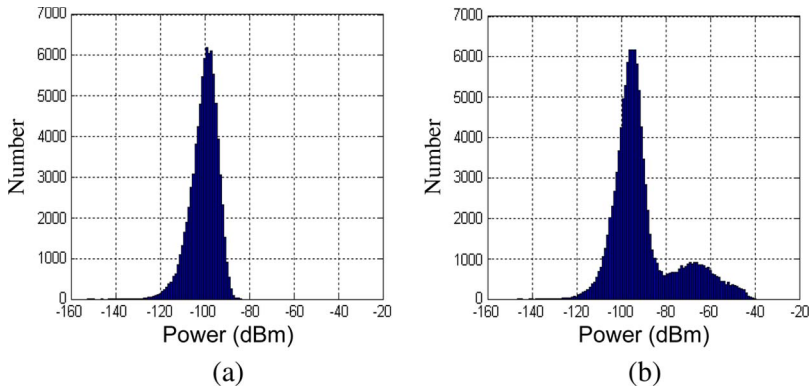


Fig. 3. (a) Histogram of the noise. (b) Histogram of the target return and noise.

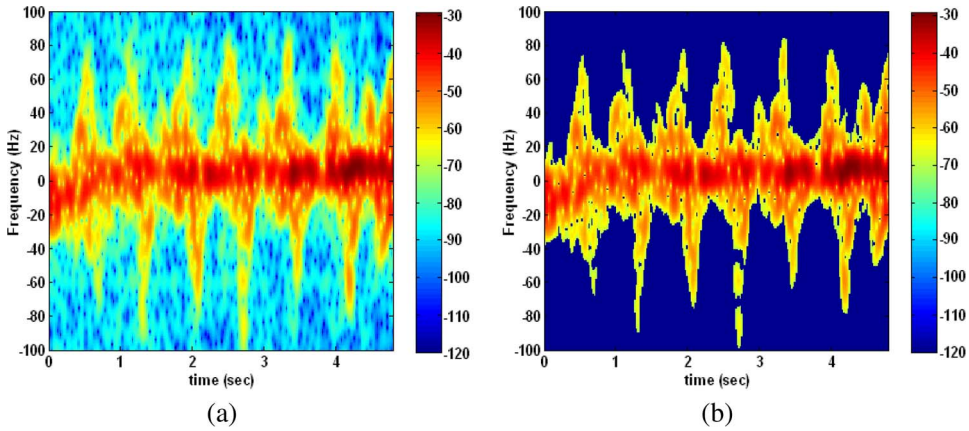


Fig. 4. Spectrogram of the boxing while standing in place motion. (a) Before the processing. (b) After thresholding.

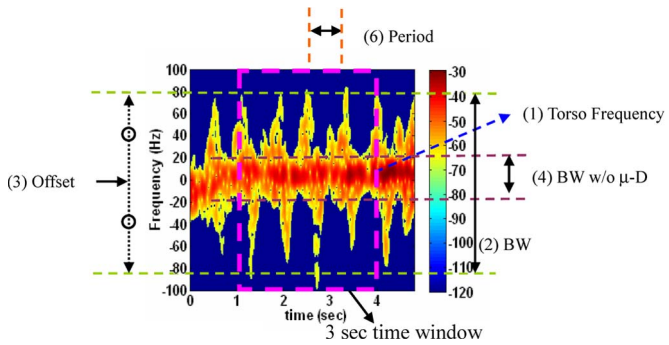


Fig. 5. Selected features of the Doppler spectrogram.

can reduce data dimensions while maintaining the essential characteristics of the micro-Doppler signatures. Based on the initial observations in Section II, we choose the following six features to characterize the micro-Doppler signatures: (1) the torso Doppler frequency, (2) the total BW of the Doppler signal, (3) the offset of the total Doppler, (4) the BW without micro-Dopplers, (5) the normalized standard deviation (STD) of the Doppler signal strength, and (6) the period of the limb motion. An illustration of the features is shown in Fig. 5.

The torso Doppler frequency (1) corresponds to the speed of the human subject. The human torso speed is a very basic yet important piece of information since there is a large variation in the torso speed depending on the different activities. The total BW of the Doppler signal (2) is related to the speed of the limb motions. A large BW results from fast swings of arms or legs. The offset of the total Doppler signal (3) is a measure of the asymmetry between the forward and backward motions of the limbs. This offset would be the same as the torso speed if the forward and backward swings are exactly symmetrical when the limbs move back and forth. The BW without micro-Dopplers (4) represents the Doppler BW of the torso alone. It accounts for the bobbing motion of the torso while the human performs an activity. The normalized STD of the Doppler signal strength (5) is related to the dynamic range of the motion. For example, a large motion like running tends to have a high STD in the Doppler signal strength, while sitting still has a small value. We normalize the STD value by the mean of the Doppler signal strength to eliminate the effect of distance between the human subject and the radar. The period of the limb motion (6) corresponds to the swing rate of the arms and legs.

A Matlab routine is written to extract these six features automatically by processing data over a 3-s time window. The torso Doppler frequency (1) is the average frequency of the peak signal in strength over the time bins within the window. For the calculation of features (2), (3), (4), and (6), two signal envelopes are first identified. The highest Doppler frequency at each time bin makes up a high-frequency envelope. The lowest Doppler frequency at each time bin makes up a low-frequency envelope. The BW (2) is the averaged difference between the largest frequencies from the high-frequency envelope and the smallest frequencies from the low-frequency envelope. The offset of the Doppler frequency (3) is the mean value between the largest frequencies of the high-frequency envelope and the

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TABLE II
MEAN VALUES OF THE SIX FEATURES

	Feature 1	Feature 2	Feature 3	Feature 4	Feature 5	Feature 6
Activity (a)	29.209	128.654	20.127	104.50	1.772	0.425
Activity (b)	22.430	120.159	21.157	85.341	1.725	0.528
Activity (c)	19.430	100.352	19.717	69.093	1.378	0.520
Activity (d)	6.491	63.691	13.570	39.197	1.294	0.539
Activity (e)	11.101	89.205	17.365	60.402	1.432	0.474
Activity (f)	0.172	97.537	5.679	50.119	0.918	0.424
Activity (g)	-0.055	30.862	-0.041	23.384	0.851	0.529

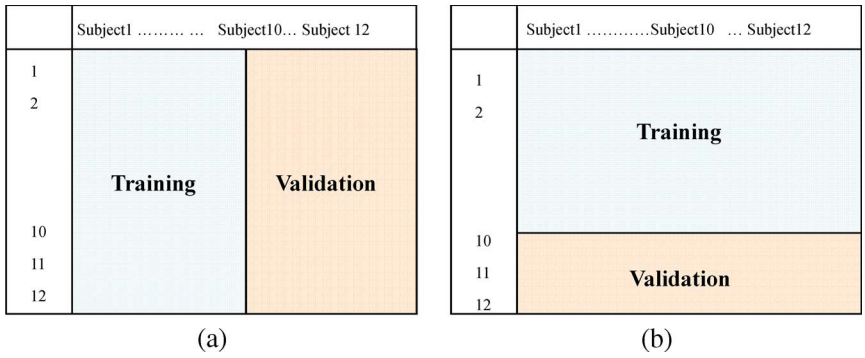


Fig. 6. Training and validation data set for the fourfold validation. (a) First scenario. (b) Second scenario.

smallest frequencies of the low-frequency envelope. The BW without micro-Dopplers (4) is the averaged difference between the smallest frequencies from the high-frequency envelope and the largest frequencies from the low-frequency envelope. Feature (5) is the STD of the signal strength divided by the average of the signal strength of all the above-noise Doppler signals in the spectrogram. The period (6) is the time period of the micro-Doppler from the limbs.

B. Training Feature Set Generation

A classifier implemented using a machine learning technique requires a training data set. In this paper, the data set is constructed by extracting the features from the measured data, which consists of 12 human subjects performing seven activities. Each human subject repeats a particular activity for 6 s four different times. Then, we use a 3-s time window to extract three different realizations of the features per measurement. Thus, 12 realizations of each activity from a particular person are formed. The total number of feature sets is $(12 \text{ subjects}) \times (7 \text{ activities}) \times (12 \text{ realizations}) = 1008$. The mean values of each feature from all 12 subjects are shown in Table II. It can be seen that the mean values show clear differences among activities, making classification potentially possible. However, by also examining the histograms (not shown here), we find that the distributions show considerable overlap with one another. This means that it is not an easy task to construct a classifier to distinguish the activities, and a more sophisticated classifier like the SVM is therefore needed.

IV. HUMAN ACTIVITY CLASSIFICATION USING SVM

An SVM is applied to classify the seven human activities based on the features extracted from the Doppler spectrogram. SVM is a binary classifier developed for non-linear boundary problems [23]–[25]. It is a supervised learning method that is based on a kernel technique. The original optimal hyperplane algorithm is limited to a linear classification problem. However, SVM is developed to create a nonlinear class boundary by applying the kernel technique to achieve the maximum margin between classes. Through the kernel technique, the input data are transformed to a high-dimensional space. In the transformed feature space, the data may be classified by a linear hyperplane. The linear optimal hyperplane results in a nonlinear boundary in the original input space. Because the classification accuracy depends on the kernel used, the selection of kernel is very critical in practice.

We utilize *LibSVM*, a freely available SVM library implemented by Chang and Lin [30]. For the given penalty parameter and the kernel parameter, *LibSVM* constructs the decision boundary based on the training data and then calculates the classification error for the validation data. In our case, three quarters of the data collected are used for the training data set, and the remaining one quarter is used as the validation set.

For the training procedure, two scenarios are used as shown in Fig. 6. The first scenario uses all the data from nine subjects for training and the data from the remaining three subjects for validation. The second scenario uses nine realizations of all

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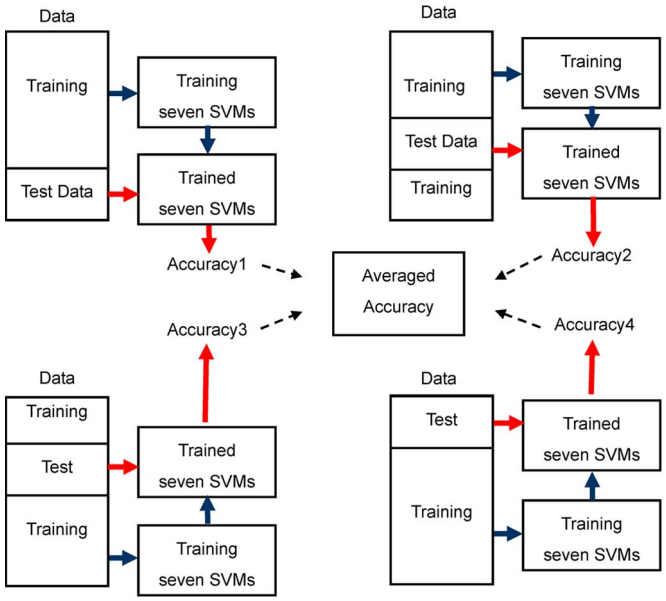


Fig. 7. Process of fourfold validation.

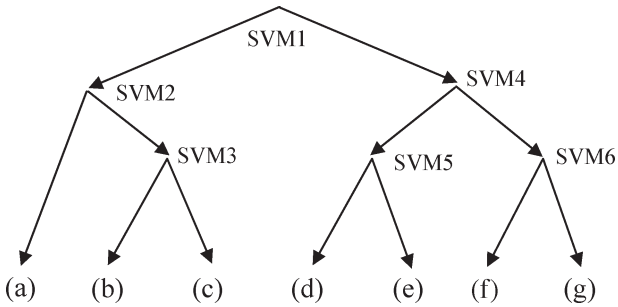


Fig. 8. Suggested decision tree using SVM.

12 subjects as the training set, and the remaining three realizations of all the subjects as the validation set. The first scenario is more realistic because it classifies the activities of unknown humans based on the data from known subjects. The second scenario classifies a person's activity based on training data that include that individual. The final classification error is calculated using a fourfold cross-validation. In the validation, the error is calculated four times with different combinations of training and validation data set for the given total data set. This process generalizes the performance of the SVM and is summarized in Fig. 7. Successful training should result in both small average error and small error variance.

The formulation of the SVM is based on the two-class problem. In order to apply an SVM to the multiclass problem like the one at hand, it should be reformulated into several binary class problems. Different methods have been developed to handle such multiclass problems using SVMs [31], [32]. In this paper, we apply SVMs to a decision-tree classifier. The decision-tree method breaks multiple classes into several distinct binary decision problems using a tree structure. At each nonleaf node of the structure, a binary SVM classifier is used. A decision-tree-based SVM is selected because it is simple and intuitive. Furthermore, it requires the least number of SVMs to be trained and has the shortest training and testing time. The configuration of the suggested decision tree is shown in Fig. 8. Six different SVMs should be constructed. Each SVM uses a different amount of training data. For example, the training data set for SVM1 consists of activities (a), (b), (c), (d), (e), (f), and (g). That of SVM3 is from only activities (b) and (c).

In our decision-tree structure, SVM1 is used at the top node to separate all activities into two groups: a group with high torso speed and a group with low torso speed. Among the group with high torso speed, the running activity (a) is separated from other two through SVM2, while SVM3 is used to classify between activities (b) and (c). SVM4, SVM5, and SVM6 are used to divide the group with low torso speed into specific activities. We note that this structure is constructed heuristically so that it may not be optimal.

We use the Gaussian kernel for the construction of the SVMs. For successful learning, the penalty parameter of the SVM and the Gaussian kernel width should be carefully determined. The optimal values are searched exhaustively to have the minimum average classification error over the fourfold cross-validation for each SVM.

For the first scenario, the average and the variance of the validation error of classification are 0.081 and 0.007, respectively. For the second scenario, the average and the variance of the validation error of classification are 0.072 and 0.010. The second scenario has slightly smaller validation error, or better accuracy (92.8% versus 91.9%), as compared to that of the first case. This is expected since the second scenario classifies a person's activity based on training data that include that individual. It is also worth noting that the performance of the SVM is better than that of an artificial neural network classifier [33], which achieved a classification accuracy in the mid-80% range.

The resulting averaged confusion matrices are shown in Table III. The boxing while moving forward (e) has the highest classification error in the first scenario, but it has a very small error in the second scenario. We believe that this is caused by the large deviations in how each subject performs this activity. Thus, predicting an unknown person's activity results in a high error. We also observe that the walking while holding a stick (c) activity is easily confused with the walking (b) activity in both scenarios. This shows that such a fine motion difference is harder to detect. In particular, at 2.4 GHz, the Doppler resolution is poor and the arm micro-Doppler is very much buried inside the micro-Doppler from the leg motions. The only distinction between the two activities appears to be the slightly smaller average values of all six features in activity (c) in comparison to activity (b), as shown earlier in Table II.

The SVM classification of activities using the Doppler features results in a low validation error. However, it is instructive to investigate which features are more important in the classification. First, we carry out the classification process using one single feature at a time. The fourfold validation error trained with only a given feature is shown in Table IV. From the table, the orders of features in terms of classification performance are as follows: the torso frequency (feature 1), the BW without micro-Dopplers (4), the total BW (2), the offset of the total

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TABLE III
RESULTS OF THE SVM BASED ON A DECISION-TREE STRUCTURE. (a) CONFUSION MATRIX FOR THE FIRST SCENARIO. (b) CONFUSION MATRIX FOR THE SECOND SCENARIO. *Es* REPRESENTS THE ESTIMATED CLASS, AND *Ac* REPRESENTS THE ACTUAL CLASS. THE NUMBER REPRESENTS THE PERCENTAGE OF CORRECT CLASSIFICATION IN EACH CASE

(a)

Es\Ac	(a)	(b)	(c)	(d)	(e)	(f)	(g)
(a)	97	0	0	0	0	0	0
(b)	0	82	17	0	2	0	0
(c)	3	18	81	0	4	0	0
(d)	0	0	0	95	6	0	0
(e)	0	0	2	5	78	1	0
(f)	0	0	0	0	10	99	0
(g)	0	0	0	0	0	0	100

(b)

Es\Ac	(a)	(b)	(c)	(d)	(e)	(f)	(g)
(a)	95	0	0	0	0	0	0
(b)	1	83	15	0	0	0	0
(c)	2	17	85	0	3	0	0
(d)	0	0	0	89	1	0	0
(e)	0	0	0	11	96	0	0
(f)	2	0	0	0	0	100	0
(g)	0	0	0	0	0	0	100

TABLE IV
CLASSIFICATION ERROR ONLY WITH A PARTICULAR FEATURE

Particular Feature	(1)	(2)	(3)	(4)	(5)	(6)
Error (Accuracy)	0.299 (70.1%)	0.456 (54.4%)	0.552 (44.8%)	0.350 (65.0%)	0.682 (31.8%)	0.697 (30.3%)

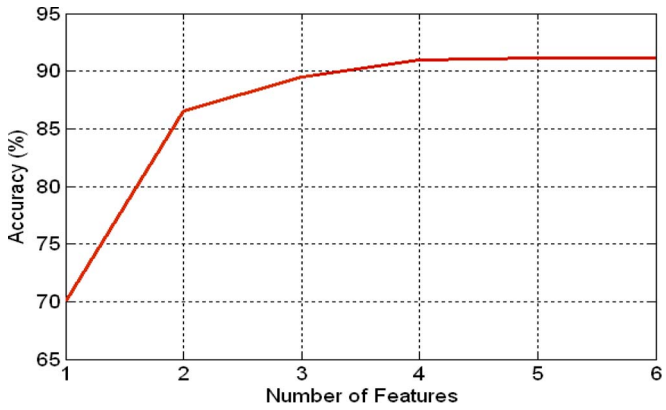


Fig. 9. Accuracy of the SVM classification with an increasing number of input features.

Doppler (3), the normalized STD (5), and the period (6). The torso frequency alone can achieve a 70% performance, while the classification accuracies from the normalized STD of signal strengths and the period are only around 30%.

Next, we evaluate the combined effect of these features by adding the features one at a time based on the orders determined earlier. Note that since the features can interact in a complex way in the classification process, it is not easy to predict the exact effect of combining features. The classification accuracy as more features are added sequentially is shown in Fig. 9. As expected, more features result in a higher accuracy overall. However, the last two features, namely, the normalized STD of signal strengths and the period, only improve the result very marginally. Therefore, good accuracy can be achieved with only the four most important features.

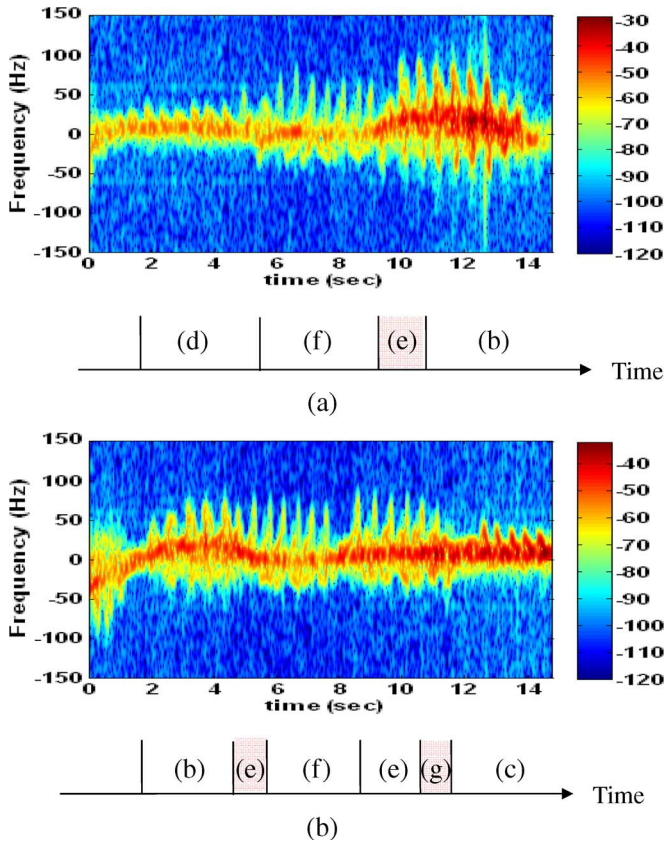


Fig. 10. Spectrograms of a sequence of activities and their classification results. (a) Crawling, boxing while standing in place, followed by walking. (b) Walking, boxing while standing in place, boxing while moving forward, followed by walking while holding a stick.

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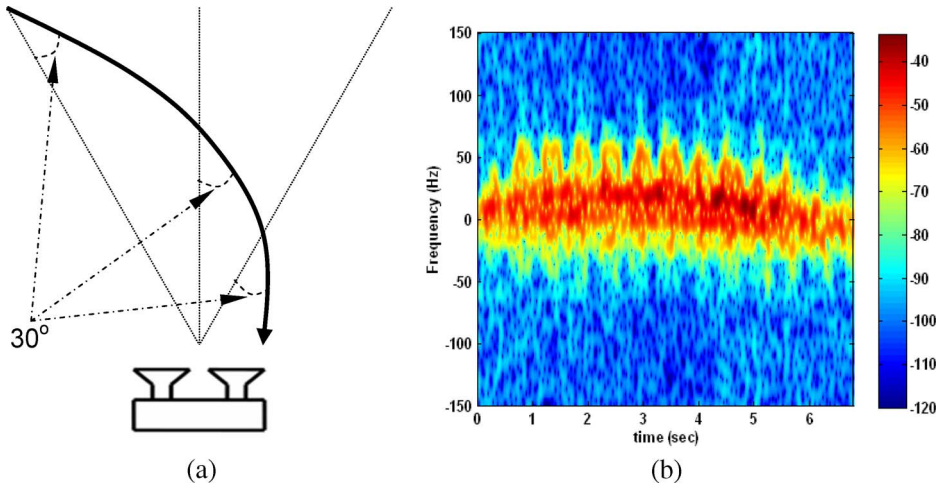


Fig. 11. (a) Trajectory of the oblique angle (30°) case. (b) Measured spectrogram for the walking motion.

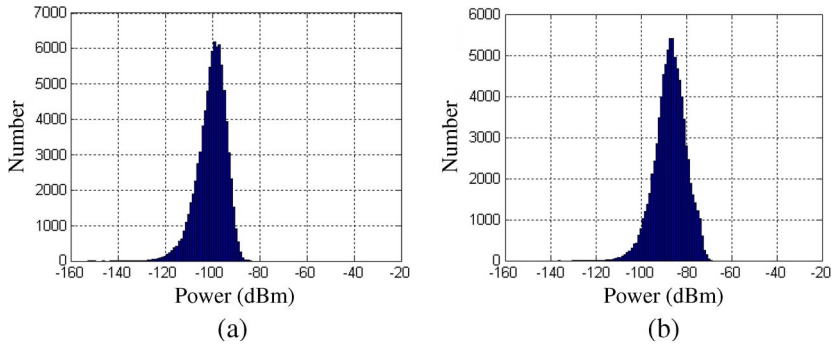


Fig. 12. (a) Histogram of noise in case of line of sight. (b) Histogram of the through-wall case.

V. Additional Investigations

A. Classification of a Sequence of Activities

In practice, human behaviors may consist of a sequence of activities when observed over an extended time period, transitioning from one activity to another. It would be useful to investigate whether it is possible to reliably classify human activities on a continuous basis. If so, then it may be possible to determine higher order human intent through the classification of a series of simple activities discussed previously. With the SVM classifier, we have developed, for single-event activities, one question that is how it will perform during the transition between activities. Measurement data from a person undergoing a sequence of activities over a longer time duration are collected and tested using the constructed SVM. A 3-s time window is first applied to the data and input to the classifier. The window is then slid forward by 0.5 s at a time, and the process is repeated onward. Two cases are considered for the sequential activities. The first case comprises crawling motion (d), followed by boxing while standing in place (f), followed by walking (b). The second case comprises walking (b), boxing while standing in place (f), and boxing while moving forward (e), followed by walking while holding a stick (c). Fig. 10(a) and (b) shows the measured spectrograms and the classification results for the two cases. The SVM predicts the activities of the first case correctly except for the transition from boxing while standing in place to walking. The misclassification region is shaded in the figure. In the transition, the activity is classified as boxing while moving forward (e). Because the features are averaged values in the 3-s time window, the classification can be erroneous when the activity changes during the time window. However, the misclassified result of boxing while moving forward (e) is not unreasonable as it can be regarded as a combination of boxing while standing in place (f) and walking (b). The second case has two intervals where misclassifications occur. The transition from walking (b) to boxing while standing in place (f) is again misclassified as boxing while moving forward (e) by the SVM. The transition from boxing moving forward (e) to walking while holding a stick (c) is classified as sitting still (g). Since these transitional motions are not included in the training data, the results for these intervals can be interpreted as examples of how our classifier performs under an unknown activity. With its structure, the classifier simply forces a decision based on the match in the extracted features.

B. Oblique Angle Case

In the collected data, the human subject always moves directly toward the radar. This will not, in general, be true in a more realistic situation. To assess the effect of the oblique

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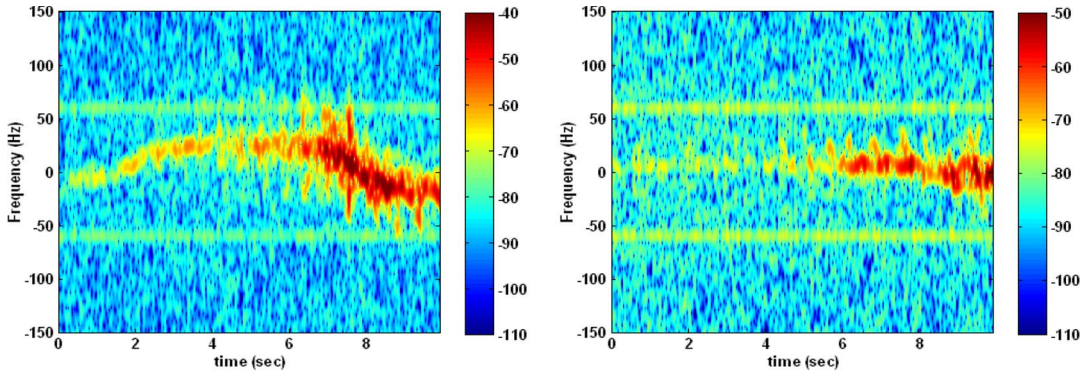


Fig. 13. (a) Through-wall measurement of the walking activity. (b) Through-wall measurement of the crawling activity.

angle case, the seven activities of a particular human subject are measured when the subject approaches to the radar while maintaining a 30° angle with respect to the radial direction. The measurement setup and the path are shown in Fig. 11(a). Note that to maintain a fixed 30° angle with respect to the radial direction, the chosen path is not straight. The subject repeats the seven activities four times. Features from 12 realizations are extracted in the same manner as before for each activity. The previously trained SVM based on the head-on data under the second scenario is then used to classify the newly measured data. The result shows almost the same classification performance. The classification accuracy was 91.4% (versus 92.8%). It is believed that the accuracy is not significantly deteriorated because the effect of the oblique-angle case is just a scaling of the Doppler information and the amount is not large when the angle is not too large. Fig. 11(b) shows a spectrogram from the walking case. Very little difference exists between this figure and the head-on walking data shown in Fig. 2(b).

C. Through-Wall Measurements

In-situ through-wall measurements are also performed to see how the micro-Doppler features are affected by wall. The test is carried out on an exterior building wall, which is a 40-cm-thick brick wall. The radar is placed in the interior of the building 0.1 m behind the wall, and a human subject carries out the activities on the other side of wall outside the building. The signal experiences approximately a 25-dB two-way attenuation due to the wall. Moreover, the noise level is increased by 11 dB due to the reflection of the incident signal from the wall. Thus, the signal-to-noise ratio (SNR) suffers a 36-dB reduction. The histograms of noise signal strength for the line of sight and through-wall cases are shown in Fig. 12. It would be much more difficult to detect the target from noise due to the reduced SNR.

Fig. 13 shows the spectrograms of the walking and the crawling activity measured in the through-wall environment. A 60-Hz ac line can now be seen due to the much lower signal level. The human subject approaches the radar as time progresses. While the torso return can still be seen, it is now much more difficult to identify the micro-Dopplers when the human subject is far from the radar (more than 4 m). When the human is within 4 m from the wall, the micro-Doppler signatures of each activity can be better observed, although the limited SNR is still troublesome for our current feature extractor. While this through-wall result is somewhat pessimistic, we believe that the main bottleneck is in our radar hardware. With a pulsed waveform and more sophisticated timing to avoid transmitter jamming, a significant improvement in the SNR may be possible. Other than the SNR issue, our recent study has shown that the wall effect on the Doppler frequency is actually quite negligible [30].

VI. Conclusion

In this paper, we investigated the feasibility of classifying different human activities based on micro-Doppler signatures. Measured data of 12 human subjects performing seven different activities were collected using a Doppler radar. Six features were extracted from the Doppler spectrogram. An SVM was then trained using the measurement features to classify the activities. The multiclass classification was implemented using a decision-tree structure. The classification accuracy based on the six features was found to be above 90%.

While the results shown in the this paper are promising, we also briefly investigated the possibility of classifying human activities over an extended time duration, at oblique angles with respect to the radar, and through a wall. Among the three issues, through-wall human classification remains the most challenging problem. The low SNR in our measurements made the use of micro-Doppler signatures very difficult. Improved hardware is needed to overcome the transmitter jamming issue. Additional studies of the wall phenomenology are also needed to understand the effect of different walls on the micro-Doppler features.

We found that a 30° oblique approach angle with respect to the radar did not significantly impact the classification results. However, more severe degradation may occur at larger angles. It may be necessary to consider distributed Doppler sensors to classify the activities of human moving in any direction.

Finally, the classification of a sequence of human activities when observed over an extended period of time was investigated. By using a moving window approach, we were able to use the developed classifier to successfully classify activities on a continuous basis. This approach may be a good first step toward determining the intent of a human by a sequence of basic activities.

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